BASE CHANGE OF THE DIRECT IMAGE ALONG A CARTESIAN SQUARE OF THE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; both steps are inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Localization of a site at an object and the associated direct image functors. Same chapter, the localization morphisms j_{X/Y} attached to the objects of a site and the description of the corresponding direct images. Supplies the two sides of the asserted identity.
- Cartesian square in a site and the induced comparison of localizations. Same chapter, the identification of the localizations of the square and the compatibility of the functors involved. Supplies the comparison making the two composites equal.

Source reference. Original lemma 08FI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4314–4314. Navigation pointer invoked at proof line 4668; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EYV: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4673–4673. Navigation pointer invoked at proof line 4673; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let*

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array}$$

be a cartesian diagram of $\mathcal{C}$. Then we have $j_{Y'/Y}^{-1} \circ Rj_{X/Y,*} = Rj_{X'/Y',*} \circ j_{X'/X}^{-1}$ as functors $D(\mathcal{C}/X) \to D(\mathcal{C}/Y')$.

Proof. Let $E \in D(\mathcal{C}/X)$. Choose a K-injective complex $\mathcal{I}^\bullet$ of abelian sheaves on $\mathcal{C}/X$ representing E. By Lemma 08FI we see that $j_{X'/X}^{-1}\mathcal{I}^\bullet$ is K-injective too. Hence we may compute $Rj_{X'/Y'}(j_{X'/X}^{-1}E)$ by $j_{X'/Y',*}j_{X'/X}^{-1}\mathcal{I}^\bullet$. Thus we see that the equality holds by Sites, Lemma 0EYV. □